



A Radical Embodied Account of Responsibility

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Abstract

In this paper, we argue that radical embodied cognitive science implies an ethics of responsibility that prioritizes what we refer to as taking collective responsibility. By *taking* responsibility, we mean that the ethical concept of responsibility ought to be more fundamentally understood in terms of the knowledge-how of the first-person capacity to respond to a situation rather than the knowledge-that of a third-person judgment that an agent ought to respond to a situation. By taking *collective* responsibility, we mean that all individual human response-taking (or, response-ability) is, to some extent, a collective endeavor. Consequently, we argue that radical embodied cognitive science has a similarly radical consequence for how we ought to think about the ethics of responsibility. To demonstrate, we consider social harms associated with use of social media as a case study to demonstrate what we mean by taking collective responsibility. In short, identifying who or what to hold responsible for social media harms obfuscates the more critical question – how do we each, as individuals, take responsibility for a phenomenon in which we have (nearly) all participated?

Keywords Radical embodied cognitive science · Collective responsibility · Group agency · Structural injustice

1 Introduction

Radical embodied cognitive science aims to reconceptualize how we understand perception, cognition, agency, and social interaction. It is radical in, at least, two ways: (1) it breaks from the representational assumptions of classical cognitive science, and (2) it aims to return to the root of cognition and agency by situating the cognizing agent within its ecological and social world. In this paper, we argue that radical embodied cognitive science has similarly radical implications for how we ought to understand moral responsibility. First, by rejecting the explanatory value of determinate representational content ‘inside the head,’ radical embodied cognitive science challenges a key presupposition of classical accounts of moral responsibility, namely that a moral

agent may be held responsible for those actions that follow from a determinate cognitive process under the agent’s control. Second, in returning the cognizing agent to its roots in its ecological and social worlds, radical embodied cognitive science suggests that our concept of responsibility ought to be similarly radicalized by returning the concept to its roots in an agent’s capacity to respond. Specifically, we argue that radical embodied cognitive science implies an ethics of responsibility that prioritizes what we refer to as taking collective responsibility. By *taking* responsibility, we mean that the ethical concept of responsibility ought to be more fundamentally understood in terms of the knowledge-how of the first-person capacity to respond to a situation rather than the knowledge-that of a third-person judgment that another ought to respond to a situation. By taking *collective* responsibility, we mean that all individual human response-taking (or, response-ability) is, to some extent, a collective and participatory endeavor.

We begin by considering traditional ways in which social ontologists have understood collective responsibility. We identify a tendency in this literature to presuppose that a group may only be held responsible insofar as the group has a determinate and institutionally codified deliberative process. Such a requirement, however, relies on a cognitivist analogy with the classical and representational model of

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understanding individual agency. This leads to two problems. First, it is inconsistent with the basic tenets of radical embodied cognitive science. Second, it fails to develop an account of responsibility in the context of groups that do not have a determinate and institutionally codified deliberative process. With this tension in mind, we then turn towards an exposition of radical embodied cognitive science. In particular, we argue that collectivity is better understood as the norm rather than the exception in radical embodied cognitive science. This is primarily due to two reasons. First, an agent's cognition and action are deeply embedded in both ecological and social processes, such that individual action (among social creatures like humans) is typically always the product of interactive dynamics between one agent and others. Second, an individual's capacity is a consequence of a collective of interactive internal components, which suggests that even the paradigmatic individual is to some extent a collective. Finally, we turn towards our positive proposal. We consider literature from both radical embodied proponents concerned with ethics and social philosophers concerned with limits of our classical conceptions of responsibility. We show an underlying commonality insofar as both reject the individualistic assumptions underlying our traditional conception of holding a person responsible. We conclude by considering the work of Donna Haraway and Karen Barad. These philosophers have attempted to articulate what we take to be a sufficiently radical reconceptualization of the ethical concept of responsibility.

2 Challenging Collective Responsibility

Consider the variety of societal harms that have been associated with social media use: brainrot, fear-of-missing-out (FOMO), addiction, depression, anxiety, attention deficits, exacerbation of political polarization, exploitative use of data acquired, and so on. Think of who or what is responsible for these harms. Perhaps what first comes to mind are the various corporate agents that have created these technologies, such as Meta (Facebook and Instagram), ByteDance (TikTok), or xAI (X, formerly known as Twitter). And, within the context of such corporate agents, we might consider whether an engineering team or an executive board or some other sub-agency holds the brunt of the responsibility. Alternatively, we might think of governmental agencies that have the capacity to regulate these technologies. And as AI systems become increasingly reason-responsive, we may at least consider the possibility of holding artificial agents accountable even if we never think such a thing can be capable of genuine moral responsibility in the sense of being praise- and blame-worthy.

Although we do not intend to diminish the significance of holding such corporate agents responsible for social harms they may have caused, we believe that this only captures part of the story. We might also think that some, or perhaps the majority, of the blame falls, quite literally, in the hands of the creators and consumers of social media themselves. The problem is that, if we think about holding users responsible, we seem to be left with one of two options: either we identify specific groups that have some kind of determinate organizational structure despite varying levels of institutional centralization and legitimacy (such as The Daily Wire, QAnon, or Russia's Internet Research Agency), or we identify individual users. Concerning the latter option, holding individuals responsible makes sense within close-knit social settings, e.g., attempting to convince your friend to put their phone away at dinner by citing harms caused to your relationship. But most of the harms associated with social media are broad social problems that seemingly cannot be solved solely through holding each other individually responsible within our close-knit relationships. Convincing your partner, for example, to stop watching conspiratorial YouTube videos doesn't address the collection of people reinforcing such content, nor does it hold your partner responsible as a member of such a collective. Although an individual, say Alex Jones, may hold some degree of personal responsibility for harms caused by the content they share online, removing such a person from virtual spaces and holding them individually responsible only addresses part of the problem. And the problem with the former option is that many of the groups associated with social media harm may be decentralized with fuzzy set membership, e.g. QAnon. If we restrict ourselves to only those with some degree of institutional legitimacy and relatively clear set membership, we preclude holding many different collections of people interacting in highly interdependent ways collectively responsible. But how do we hold distributed collectives with fuzzy set membership and no institutional legitimacy responsible? Is it even philosophically coherent? And is it something we ought to do?

To address these questions, we must first ask—what justifies holding a group responsible in the first place? Individual humans are self-conscious persons capable of understanding why they acted, communicating that understanding, and changing their behavior in light of that understanding. At least on first glance, groups are not self-conscious persons. French (1979), however, argues that corporate agents are indeed persons. He argues that, because corporate agents are capable of acting intentionally, responding to reasons, and altering behavior in light of those reasons, corporate agents may indeed be recognized as moral persons. List and Pettit (2011), however, take a different approach. They argue that being a moral agent capable of being held responsible,

although a kind of personhood, is not the same as the intrinsic personhood ascribed to all human beings. Being a moral person, on their view, is performative. Both groups and adult humans are recognized as moral persons insofar as they have the capacity to act intentionally, respond to reasons, and alter their behavior. Although infants are recognized as intrinsic persons, they are not capable of the performativity required for moral personhood. This is something that must be taught. In this way, List and Pettit suggest that certain kinds of groups can satisfy the requirement of being moral agents insofar as they are capable of performing the role of a reason-responsive person in relation to other groups.

What unites both of these accounts of collective responsibility is that a group may be held responsible insofar as that group is reason-responsive. Facebook executives deliberated the general structure of the technology, engineering teams deliberated how to design the technology, marketing and ethics teams deliberated how to monitor, filter, and censor content, the Russian government deliberated how to instruct the Internet Research Agency in its online propaganda efforts, and the US government deliberated what kinds of laws and regulations ought to be implemented. Each of these group agents are capable of responding to moral and legal reasons from other groups, deliberating, and altering their future behavior as a group. Crucially, however, all of these cases involve relatively clearly defined group boundaries and institutionally recognized deliberative processes. This renders such groups reason-responsive. But what about the distributed mass of social media consumers that reinforce the machine learning recommendation systems? Again, we could hold them responsible as individuals. But collective responsibility seems off the table because such collections do not have any determinate group structure and institutionally codified deliberative process rendering them, at least on first glance, incapable of responding to reasons as a group. We may be able to identify, for example, the collection of all people that have participated in reinforcing the popularity and spread of the Pizzagate conspiracy, but this collection has neither well-defined membership nor a codified deliberative process. How can we hold the collective responsible without such a determinate group structure?

Consider the analogue between a group's deliberative process and cognition. When an executive board meets, that deliberative process is internal to the group. Individuals present are typically in-group members. The meeting is socially recognized as a meeting of the group, or some sub-group constitutive of the primary group. Such determinate and institutionally codified deliberative processes are analogous to the classical conception of cognition as internal manipulation of representations. There are modules within the brain, or the 'mind,' manipulating informational representations of the agent, the agent's world, and the relationship between

the agent and its world. Actions are generated as a causal consequence of such representation-manipulation modules happening within cognition. Likewise, an executive board meeting involves internal deliberation over representations of the company, the company's social and economic world, and the relationship between the company and its world. Adopting a classical view of cognition, it is hard to ascribe agency and thus responsibility to very many groups outside of those with some kind of determinate group membership and institutionally codified deliberative process.

The connection between classical cognitive science and collective responsibility can be seen in Pettit's (2007) account of collective responsibility. In defining "corporate agents," he writes:

"Examples include companies, parties, churches, and universities but also partnerships, voluntary associations, and town meetings. These entities operate via their members, although they retain their corporate identity through continuous changes of membership. Specifically, they operate through their members in such a way that they simulate the performance of individual agents. They endorse certain goals and methods of reviewing goals and certain judgments and methods of updating judgments, and they follow procedures that enable them to pursue those goals in a manner that makes sense according to those judgments." (172).

In the case of distributed collectives with indeterminate group membership and no codified deliberative process, we can't really say that there are "procedures that enable" the collective to form, review, and pursue various goals and judgments. Pettit's emphasis, here, seems to be on groups with determinate group membership and codified deliberative processes. This is because he is relying on an agential idealization from classical cognitive science, namely that we can understand how and why an agent acts by thinking of the various propositional states that the agent either believes or desires. It's relatively straightforward to talk of the beliefs and desires of a company with an explicit mission statement, a board of directors, and so on. However, it's difficult to talk about the collective beliefs and desires of the collection of all people that have participated in popularizing the Pizzagate conspiracy.

But does the lack of a determinate and institutionally codified deliberative process actually entail that such a collective is not reason-responsive as a collective? In posing this question, we believe we are returning to the very question of whether or not a representational understanding of action is appropriate, a question that resides at the heart of radical embodied cognitive science. What makes a deliberative process determinate and institutionally codified? It

involves a group of users coming together and representing their intentions explicitly to one another while collectively determining how the group should act. Collections of social media users typically have not met one another even though their daily behavior, both online and offline, is highly interdependent upon the same social media pathways they traverse virtually. Such collectives are similar to the crowd behavior early social scientists attempted to explain. There does seem to be some genuine sense of collective action, albeit virtual, in the same way that emergent patterns of collective coordination produces crowd behavior. Such crowd behavior, whether virtual or not, may be responsive to reasons, but not in the same way as a determinate and institutionally codified deliberative process. If we reject a representational understanding of action, does this undermine the underlying cognitivist analogy in Pettit's account of collective responsibility? And if so, how ought we think about collective responsibility in the context of virtual crowd behavior on social media? In short, how do we understand collective action and responsibility when the collective in question lacks a determinate and institutionally codified deliberative process?

3 Radical Embodied Cognitive Science Meets Social Ontology

To answer these questions, we turn towards radical embodied cognitive science, enactivism, and participatory sense-making. Radical embodied cognitive science (Thompson and Varela 2001; Chemero 2009) is a philosophical and scientific approach to studying phenomena such as perception, action, cognition, and social interactions. It begins with an understanding of the relationship between mind and world as general as those by James and Eleanor Gibson and their followers, phenomenologists such as Heidegger or Merleau-Ponty, and proponents of the enactive approach in cognitive science. Each of these sets of thinkers begins with a critique of an understanding of cognition according to which the world causally impinges on the thinker and causes the thinker to form internal representations of the world, where these representations are the thinker's only access to the world. With the exception of a few decades in the twentieth century, this representationalist approach has been the scientific and lay understanding of the mind, from Descartes to today's cognitive and neural sciences. In contrast, in the world that the Gibsons, phenomenologists, and enactivists depict, a thinker is surrounded by and interacting with the world itself and need not form representations of it to do so.

The recent interest in dynamical systems theory in the cognitive sciences begins with work by Kugler et al. (1980) in their attempt to answer a question raised by James

Gibson. In trying to explain action in a way that did not demand an inner agent using sensory representations of the world to develop motor representations of the actions to be taken, Gibson says that "The rules that govern behavior are not like laws enforced by an authority or decisions made by a commander: behavior is regular without being regulated. The question is how this can be" (Gibson 1979). Kugler, Kelso and Turvey answered this question by arguing that human behavior is self-organizing, and therefore ought to be treated using dynamical systems theory, which is the set of mathematical tools applied to self-organizing systems in other sciences. A dynamical system is a set of variables changing continually, concurrently, and interdependently over time in accordance with some dynamical law. In principle, if not always in practice, these laws can be described as a set of differential equations which model how the variables change over time. Dynamical systems theory is especially appropriate for phenomenological, ecological, and enactive approaches because single dynamical systems can have parameters and variables both within the agent and in the environment or another agent.

The use of dynamical systems theory as a modeling tool plays several crucial roles in radical embodied cognitive science. First, and perhaps most importantly, it bridges the gaps between abstract theorizing and concrete data that can be gathered in the lab. Secondly, phenomenological, ecological, and enactivist approaches do not understand cognition primarily in terms of representation. They demand an explanatory tool that can span the agent–environment border. Dynamical systems explanations do this. Radical embodied cognitive science, then, is the use of dynamical systems theory to put the theoretical positions of phenomenology and the ecological and enactive approaches in touch with data about perception, action, and cognition that can be gathered in the lab.

The centrality of social interaction has been a key feature of radical embodied cognitive science from the beginning. As noted above, dynamical systems theory can equally easily model activity within a biological organism and across organisms. Schmidt et al. (1990) showed that while two people swing their limbs in synchrony, they are connected to one another in the same way that the limbs of a single individual are connected. That is, the interpersonal connection between two people is a temporary version of the intrapersonal connection among the parts of a single person. This has been extended by Gibsonian psychologists to many different kinds of social interaction. Richardson et al. (2007) showed that participants seated next to one another on rocking chairs behave just as the coordinating limbs of individual humans; Patrick Nalepka and colleagues (2017, 2019) did the same for pairs of participants who played a two-person video game.

De Jaegher and Di Paolo (2007) extend the enactive approach to social interactions, via what they call *participatory sense-making*. In participatory sense-making, two individuals are coupled with the world and with one another, such that they collectively and temporarily open a new domain of significant interactions which is not available to either separately. To take an example from De Jaegher and Di Paolo, consider what happens in what we might call ‘the hallway dance’, when each of two individual humans attempt to make space for the other while passing in a narrow hallway. Each of these individuals is an agent. Given their interest in avoiding collision with one another, which would have negative consequences for both, each will move to one side of the hallway to let the other pass. Most of the time this works perfectly well, but sometimes both individuals will move to the same side of the hallway, yielding another potential collision. Because each experiences potential collisions negatively, they each simultaneously then move to the opposite side of the narrow hallway, setting up another potential collision. This can happen several times, with each person mirroring the other’s movement repeatedly. This hallway dance is an example of participatory sense-making. In it, each individual remains an agent, even while they are temporarily components of a group agent. Notice that, like the individuals who participate in it, the hallway dance displays a kind of autonomy (Fuchs and De Jaegher 2009). The dance maintains itself as an entity, at least for a while, because each attempt at collision avoidance by the participants leads to another possible collision, the avoidance of which leads to another possible collision, and so on.

Across radical embodied cognitive science, groups of individuals in social coordination are treated in just the same way as the coordinating components of a single agent. If an individual biological organism is an agent, so too are coordinating groups of individual biological organisms. And because individual human action nearly always happens within the context of a social world constructed through participatory sense-making with others, the causal responsibility for individual action is nearly always distributed throughout various collectives of which an individual is a constituent. Moreover, because individual action is typically always bound by social relations and is performative in some way, each instance of individual action is not only an expression of one collective agent but typically multiple. Consider our writing this paper. We are acting on behalf of ourselves as individuals with academic careers, as a pair of two academic colleagues, as proponents of radical embodied cognitive science, as philosophers, and as global citizens concerned about how certain technologies are used, just to name a few. A key tenet of participatory sense-making is that we can, at most, participate in the collective act of all of

these collectivities of which we are a part; we cannot and do not have absolute control over any such collective action. In this way, we claim that group agency is the norm in radical embodied cognitive science.

That group agency is, or at least ought to be, the norm in radical embodied cognitive science is perhaps best demonstrated by a tension that lies at the core of radical embodied, ecological, and enactive appeals to biology to explain cognition. In general, 4E accounts of cognition place considerable emphasis on *individual* organisms. Enactivism and ecological psychology, for example, emphasizes the relationship between the individual organism and its environment, while embodied cognitive science emphasizes the individual organism’s body in explaining that organism’s cognition. Corris (2023) points out, however, that this emphasis on the individual organism is in tension with 4E cognitive science’s general appeal to recent biological accounts that emphasize that organic development involves the interaction of distributed assemblages. Various authors have argued that, in fact, collectivity ought to be understood as the biological norm rather than the exception (Falandays 2023; Levin 2024; McMillen and Levin 2024; Watson and Levin 2023). Dengsø and Kirchoff (2023) argue that, consequently, 4E cognitive science ought not prioritize individual autonomy and instead emphasize “dynamic differentiation” (351).

Much of this emphasis on collectivity and dynamic differentiation comes from the work of Lynn Margulis. She argued that the traditional conception of the organism ought to be replaced with what she and others have referred to as a *holobiont*. In some sense, a holobiont is a concept somewhere in between the traditional conception of an organism and the concept of an ecosystem. Our traditional concept of an organism suggests that an organism is a discrete individual composed of genetically identical cellular constituents. However, as we have come to understand the complexities of microbiotic processes both within and outside of the body, we have learned that much of the biomass of an organism consists of genetically dissimilar cellular constituents. The concept of symbiosis is not sufficient to capture the interactions between a host and the microorganisms that live within and on that host. This is because, without these microbiotic processes, the host organism could not persist. Consequently, Margulis and others push us to recognize the holobiont (host+microorganic symbionts) as a collective and dynamically differentiated system that acts as the proper unit of life to be studied within biology and evolutionary theory. From this point of view, what we have traditionally thought of as individuals are, in fact, groups.

Citing the master’s thesis of Dempster (1998), Haraway (2016) argues that sympoiesis, rather than autopoiesis as proposed by enactive theorists, is the defining feature of

living systems. Autopoiesis (“self-making”) emphasizes that life involves systems that create and maintain their own boundaries, while sympoiesis (“making-with”) emphasizes that life is always a process of collective coordination and dynamic differentiation. In attempt to dispel the tension pointed out by Corris (2023), Dengsø and Kirchoff (2023) argue that sympoiesis ought to be the basis for 4E cognitive science, insofar as cognition is both the product of dynamically differentiated, constituent processes collectively coordinating and is intrinsically social as pointed out by the account of participatory sense-making developed by De Jaegher and Di Paolo (2007).

In sum, not only is it the case that the very worlds in which we participate are socially co-created such that responsibility is distributed, and not only are our actions mostly expressions of various collective agents of which we are apart, but, additionally, our own individual actions are better understood as collective actions of the various sympoietic webs within which we find ourselves. We are, as individuals, both constituted by and constituents of collectives. Thus, the ethical questions of *collective* action and responsibility, for radical embodied cognitive science, are themselves the primordial questions of action and responsibility in general.

4 Taking Collective Responsibility

Proponents of radical embodied cognitive science typically agree that the theoretical perspective is inherently normative, or at least has direct normative implications. Jayawickreme and Chemero (2008) argue that moral affordances can be investigated in a way that is simultaneously both scientific and value-laden thereby justifying a form of moral realism grounded in radical embodied cognitive science. Colombetti and Torrance (2009) argue that an enactivist approach to ethics should shift emphasis away from the autonomy of individual agents and towards the social interaction between them. Moreover, they suggest that participatory sense-making is intrinsically normative, such that all social interactions are value-laden. In this way, an enactive ethics aligns with the ethical theorizing of philosophers such as Buber and Levinas that place emphasis on the intrinsically ethical character of encountering the other. Fourlas and Cuffari (2022) claim that ethics is not an “add-on” to enactivism as “there are no morally neutral circumstances for complex living beings” (357). Extending an argument developed by Colombetti and Torrance, they argue that, because enactivism challenges the individualism implicit in traditional conceptions of moral responsibility, practically addressing ethical problems such as structural racism requires a sensitivity to how one interacts with and

contributes to one’s community and cannot be addressed solely by blaming wrongful actors. Because our social value systems are being co-created through dynamics described by participatory sense-making, responsibility for structural injustices is distributed.

A similar critique of the applicability of traditional accounts of moral responsibility to structure injustice may also be found in the work of social philosophers. For instance, Zheng (2021) suggests that the traditional framework of holding specific agents responsible breaks down as an adequate response to problems such as structural injustice. Even if individual actions participate in perpetuating social injustice, the consequence of perpetuating the injustice may be unintentional and the individual likely has little to no power to stop the injustice on their own. In this spirit, Haslanger (2019) expresses concern that failing to identify how social processes shape the way we think thereby allows ideologies, some of which may be unethical and potentially dangerous, to overdetermine how we think. For both views expressed in this paper and others, Haslanger has been criticized for failing to adequately give attention to how individual agency can resist such ideological oppression. Maiese (2022), however, turns towards enactivism as a framework that explains how our social participation shapes our mind in both constructive and oppressive ways. Citing the enactive conception of participatory sense-making, she places emphasis on how the heteronomy of social norms, although potentially posing a threat of oppression, also enables individuals by extending their agency. What’s crucial in her framework, for our purposes, is the extent to which she agrees with a philosopher like Zheng. In short, when faced with structural injustice and ideological oppression, the most important ethical question to ask oneself is – how do I respond? – and not – who or what ought to be held responsible?

Zheng (2021) argues that backwards-looking attempts to hold agents responsible for structural injustice are misguided. This doesn’t mean there’s no responsibility to go around in the case of structural injustice. Rather, emphasis ought to be forward-looking and focused on taking responsibility with others. In other words, it’s not the case that there’s no responsibility for injustice. It’s just not the kind of responsibility in which individuals can be blamed. The burden lies with us all. She differentiates between “imperative” and “aspirational” modes of thinking about moral agency. According to the first mode, agents are free to deliberate and choose their own actions. Consequently, some types of moral demands are placed on agents such that specific types of actions are deemed immoral. In such cases, we may blame an agent for wrongdoing and ask them to take responsibility for the alleged wrong. According to the second mode, agents’ freedom is often quite limited. As

Zheng puts it, “we do not experience things as fully up to us, but we feel called upon to do something” (505). In such cases, we may aspire to take responsibility for some moral problem even without being blameworthy. Zheng aims to address such structural injustice by developing an account of moral criticism aimed at the aspirational goal of taking responsibility for injustices for which one cannot be held responsible.

Similarly, Conradie and Nagel (2025) develop an account of taking extended responsibility in the context of AI ethics, wherein an individual takes responsibility for something for which they – and perhaps no determinate agent – can reasonably be held responsible. As they see it, such an approach to AI ethics is unavoidable as general-purpose, semi-autonomous AI systems increasingly distribute responsibility for AI-induced harms across many people. Thus, they consider an emphasis on first-person response-taking to be a necessary response to the problem of many hands that intrinsically characterizes general-purpose systems trained on large datasets of collectively created information.

It is precisely this perspective that we argue applies to cases of collective action when the collection in question has indeterminate membership and no codified deliberative process, such as many of the harms caused by collective behavior on social media. We think we are all responsible in the forward-looking sense to take responsibility for these harms because, in a very important sense, we are all causally responsible for the harms in the first place. Crucially, however, we argue that this perspective follows from the key tenets of radical embodied cognitive science as we have described them, such that taking collective responsibility is not merely the proper perspective to take in the context of structural injustices but in nearly all cases of ethical action. As we have argued, participatory sense-making and considerations from biology suggest that all individual action is collective.

The traditional debate concerning moral responsibility tends to focus on attribution of responsibility by a third party. This third party deems an agent either blameworthy or praiseworthy for an action. Whether blame or praise is attributed depends on the intentions, i.e., mental representations, of an agent and on the moral value of the act and/or its consequences. In both cases, responsibility is something attributed to an agent from the outside. The traditional ethical debate concerns whether or not this attribution is justifiable. In this sense, moral responsibility is separated from the agent. This is similar to how the traditional approach of Descartes and contemporary cognitive scientists separate cognizing from the rest of the features of an agent’s embodiment. For Descartes, this separation is an ontological rift between the mind and the material world and leads to the positing of a special mental realm. In the hands of

materialist cognitive scientists, this separation becomes a distinction between the world outside the brain and the representation of the world inside the brain.

What’s lost in the separation between world and representation in the brain is the presence of the world (and the body) in cognition. Similarly, what’s lost in the separation between moral responsibility (conceptualized as an attribute and dependent on the attribution of mentally represented intentions) and moral agency is the capacity of the moral agent in question to respond to its situation. Donna Haraway and Karen Barad call the latter the “response-ability” of the moral agent (Barad 2007; Haraway 2016). In other words, an ethics of moral responsibility conceptualized as third-person attribution is like a science of the mind conceptualized as a collection of representations. Both approaches focus on a target that fails to capture the primordial significance of the thing targeted. Minded agents enact a world, and moral agents are response-able for that enactment. This is not a static attribution to the moral agent; rather, it is a dynamic capacity of the agent to engage with and change its environment. Thus, an ethics of responsibility should not prioritize questions such as – is agent A responsible for event E? – but rather – to what possibility space P can agent A actually respond?

This refocusing of our attention on taking responsibility rather than holding responsible, we believe, is more consistent with the above-mentioned argument that cognition and life is better understood as a collective process of dynamic differentiation rather than individual autonomy. What we think of as individuals are complex, and their reasons for acting are complex and often involve competing and contradictory thoughts and tendencies. Moreover, what we think of as individual actions are typically always expressions of group agencies of which the individual is a part just as much as they are expressions of the individual’s own autonomy. Consider, for example, the choice to pick up a phone and watch reels on TikTok. There is some degree of individual choice in this action, but there are also large-scale, group dynamics at play. Social media recommendation algorithms attempt to recommend content that keeps the user engaged as long as possible. Moreover, social media content creators are attempting to appeal to the interests of large collections of people. Consequently, whenever someone chooses to watch TikTok reels, their individual action is not only an expression of their own individual agency but also multiple nested and interacting group agencies of which they are a part. Consequently, identifying who is responsible for the agent’s watching TikTok reels is complicated. In general, identifying who is responsible for any given action is quite complicated. This lack of a yes-or-no answer to questions such as ‘Is agent A responsible for event E?’ is captured by Haraway’s (1991) use of the term *cyborg*. In Haraway’s

cyborgian philosophy, rigid binaries are rejected, such as the binary distinctions between human and technology, but also man and woman, culture and nature, and so on. We extend this to propose a rejection of the distinction between group and individual agents insofar as an individual agent's action is typically always an expression of their own agency and the groups of which they are a part.

We believe that rejecting the rigid distinction between group and individual agency ought to cause us to reconceptualize moral responsibility. Citing Emmanuel Levinas, Barad ends her book, *Meeting the Universe Halfway*, with what she refers to as an “ethics of mattering” (391). She writes, “Levinas rejects the metaphysics of the self that serves as a foundation for conventional approaches to ethics. Subjectivity is not a matter of individuality but a relation of responsibility to the other. Crucially, then, the ethical subject is not the disembodied rational subject of traditional ethics but rather an embodied sensibility, which responds to its proximal relationship to the other through a mode of wonderment that is antecedent to consciousness” (391). We interpret this to imply that the capacity to take responsibility, or the ability to respond (i.e., response-ability), is the more fundamental concept when we think of moral responsibility. The third-person emphasis on attributing mental representations in order to hold an agent responsible for their actions presupposes a more fundamental account of response-ability, or as we put it “taking responsibility.” Conceptualized as such, moral responsibility is not fundamentally a concept concerned with individual agency but is rather more fundamentally a concept concerned with relationality. In this sense, what Haraway and Barad call “response-ability” is deeply influenced by accounts of relational autonomy coming out of recent feminist philosophy.

To demonstrate, consider Haraway's discussion of a pigeon loft built in Batman Park in Melbourne, Australia. As she describes it, the loft was designed to maintain the pigeon population by encouraging “the pigeons to roost away from city buildings and streets” (27). She writes, “I want to see the pigeon loft as a small, practical enactment and a reminder to further opening to the response-ability of staying with the trouble. Response-ability is about both absence and presence, killing and nurturing, living and dying – and remembering who lives and who dies and how in the string figures of naturalcultural history.” (28) The significance of this passage, as we understand it, is a rejection of traditional accounts of moral responsibility and their applicability to dealing with contemporary climate crises. The tenuous relationship between urban environments and pigeon populations is wrapped into a history of capitalism, colonialism and industrial development. Consequently, we might think the proper response to this situation is to identify the agents, whether they be individual or group agents,

that ought to be held responsible. Doing so, however, is complicated, and most of the agents to be held responsible are no longer living. Moreover, the creation of these pigeon lofts was not conducted by any such historical agent(s). Rather, contemporary people living in the midst of a tenuous relationship between local pigeon populations and the urban environments inhabited by those people are taking responsibility for a state of affairs for which they ought not be held responsible.

5 Conclusion

We take the work of Haraway and Barad to be the best example of the kind of ethics of responsibility (or, response-ability) implied by radical embodied cognitive science. Consider the case of social media harm and the use of TikTok that we've been discussing throughout the paper. We could, as we considered in Sect. 2, attempt to identify the specific agents that ought to be held responsible and blameworthy for the various harms associated with social media. But, this obfuscates the more fundamental fact. Just as modern city-dwellers and pigeons must live and die side by side by no choice of their own, we are living in a generation that is compelled to engage with social media by no choice of our own. Our first priority in dealing with such harms, thus, should be identifying ways in which we participate in virtual collective agents and developing the knowledge-how and sensibility to respond in ways that increase the values that we want represented online. As radical embodied philosophers, we take this to not only be the only plausible way of addressing harms associated with social media but also the most ethically justifiable way of responding.

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Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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